Calculation of the factor si

1 The factor si shall be determined for each case of assumed flooding, involving a compartment or group of compartments, in accordance with the following notations and the provisions in this regulation.

e is the equilibrium heel angle in any stage of flooding, in degrees;

v is the angle, in any stage of fl ooding, where the righting lever becomes negative, or the angle at which an opening incapable of being closed weathertight becomes submerged;

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SOURCE: MSC.216%2882%29.pdf
v is the angle, in any stage of fl ooding, where the righting lever becomes negative, or the angle at which an opening incapable of being closed weathertight becomes submerged;

GZmax is the maximum positive righting lever, in metres, up to the angle v;
RESOLUTION MSC.216(82)
(adopted on 8 December 2006)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED-

Range is the range of positive righting levers, in degrees, measured from the angle e. The positive range is to be taken up to the angle v;

Flooding stage is any discrete step during the flooding process, including the stage before equalization (if any) until final equilibrium has been reached.

1.1 The factor si, for any damage case at any initial loading condition, di, shall be obtained from the formula:

si = minimum { sintermediate,i or sfinal,i · smom,i }

where:

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SOURCE: MSC.216%2882%29.pdf
1.1 The factor si, for any damage case at any initial loading condition, di, shall be obtained from the formula:

si = minimum { sintermediate,i or sfinal,i · smom,i }

where:

sintermediate,i is the probability to survive all intermediate flooding stages until the final equilibrium stage, and is calculated in accordance with paragraph 2;

sfinal,i is the probability to survive in the fina l equilibrium stage of flooding. It is calculated in accordance with paragraph 3;

1.
smom,i is the probability to survive heel ing moments, and is calculated in accordance with paragraph 4.

2 The factor sintermediate,i is applicable only to passenger ships (for cargo ships sintermediate,i should be taken as unity) and shall be taken as the least of the s-factors obtained from all flooding stages including the stage before equalization, if any, and is to be calculated as follows:

sintermediate,i =
4
1
max
7 05 . 0

RangeGZ

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obtained from all flooding stages including the stage before equalization, if any, and is to be calculated as follows:

sintermediate,i =
4
1
max
7 05 . 0

RangeGZ

where GZmax is not to be taken as more than 0.05 m and Range a s n o t m o r e t h a n 7 ° . sintermediate = 0, if the intermediate heel angle exceeds 15º. Where cross-flooding fittings are required, the time for equalization shall not exceed 10 min.

3 The factor sfinal,i shall be obtained from the formula:

sfinal,i =
4
1
max
16 12 . 0

RangeGZK

where:

GZmax is not to be taken as more than 0.12 m;

Range is not to be taken as more than 16°;

K = 1 if e min

K = 0 if e max

K =
minmax
max

e otherwise,

where:

min is 7° for passenger ships and 25° for cargo ships; and

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SOURCE: MSC.216%2882%29.pdf
K =
minmax
max

e otherwise,

where:

min is 7° for passenger ships and 25° for cargo ships; and

max is 15° for passenger ships and 30° for cargo ships.

4 The factor smom,i is applicable only to passenger ships (for cargo ships smom,i shall be taken as unity) and shall be calculated at the final equilibrium from the formula:

()
heel
max
mom,
M
ntDisplaceme04 . 0 GZs i =
where:

Displacement is the intact displacement at the subdivision draught;

Mheel is the maximum assumed heeling moment as calculated in accordance with paragraph 4.1; and

smom,i 1

4.1 The heeling moment Mheel is to be calculated as follows:

Mheel = maximum { Mpassenger or Mwind or MSurvivalcraft}

4.1.1 Mpassenger is the maximum assumed heeling moment resulting from movement of passengers, and is to be obtained as follows:

Mpassenger = (0.075 · Np) · (0.45 · B) (tm)

where:

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SOURCE: MSC.216%2882%29.pdf
4.1.1 Mpassenger is the maximum assumed heeling moment resulting from movement of passengers, and is to be obtained as follows:

Mpassenger = (0.075 · Np) · (0.45 · B) (tm)

where:

Np is the maximum number of passengers pe rmitted to be on board in the service condition corresponding to the deepest subdivision draught under consideration;